

## 1 Query & Contract



User Query

### Evidence Contract



Risk  $\leq \alpha$



Cost  $\downarrow$

### Plan Space $\mathcal{P}$



### Heterogeneous RAG plans

$\mathcal{P} = \{p_1, p_2, \dots, p_P\} \cup \{\perp\}$

## 2 Ledger Gate

$$B_{t-1} = \alpha(t-1) - V_{t-1}$$



IF  $B_{t-1} \geq 1 - \alpha$

**Gate OPEN** ✓

**Action LP**

(go to Step 3)



ELSE  $B_{t-1} < 1 - \alpha$

**Gate CLOSED** ✗

**Decline**  $\perp$   
with  $\ell_t = 0$



Pathwise guarantee:

$V_t \leq \alpha t$  for all  $t \leq T$

No  $\delta$ , no exchangeability

## 3 Cost Optimisation

1

Action LP

$$\min_{w \in \Delta} \sum_a w_a c_a \quad \text{s.t.} \quad \sum_a w_a \tilde{r}_a \leq \alpha$$

over  $\mathcal{P} \cup \{\perp\}$

$$\text{Optimistic risk: } \tilde{r}_a = \hat{r}_a + \frac{\kappa}{\sqrt{n_a}}$$

2



Sample action  $a \sim w$

3



Execute plan  $a \in \mathcal{P}$   
or decline  $\perp$

4



Audit loss  $\ell_t \in [0, 1]$   
(1 if evidence violates contract)

5



Update ledger

$$B_t \leftarrow B_{t-1} + \alpha - \ell_t$$

6



Learn estimates

$\hat{r}_a, c_a$  for all  $a \in \mathcal{P}$



Regret guarantee

(to best fixed plan in hindsight)

$$O\left(\sqrt{\frac{P \log T}{T}}\right)$$



### Validity by accounting

Always respect risk budget  $\alpha$   
pathwise, without exceptions.



### Cost by learning

Optimize expected cost  
while controlling risk.



### Decline as first-class action

Decline when budget is tight;  
zero loss, zero regret.